



The reinforced photothermal effect of conjugated dye/graphene oxide-based phase change materials: Fluorescence resonance energy transfer and applications in solar-thermal energy storage

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ABSTRACT

The development of shape-stabilized phase change materials (ss-PCMs) with efficient solar energy conversion performance, large energy storage capacity, and high thermal conductivity is essential for solar energy utilization. Here, novel photo-driven composite ss-PCMs were successfully fabricated by grafting blue anthraquinone dyes (Bdye) on carboxylated graphene oxide (GO) and impregnating poly(ethylene glycol) (PEG) into Bdye-grafted GO nanoconjugate system (GO-co-Bdye). The connection between grafting Bdye and GO was close enough for fluorescence resonance energy transfer from Bdye to GO. The GO-co-Bdye with a three-dimensional (3D) interconnected network structure prevented the leakage of PEG above its melting point, improved the crystalline nature, thermal stability, and thermal conductivity of composite PCMs. The composite displayed an excellent photothermal conversion capacity of 86.6% and improved thermal conductivity of 98.0% compared with the pristine PEG. The thermogravimetric analysis and differential scanning calorimetry showed that the composite PCMs exhibit high thermal stability and thermal storage performance.

1. Introduction

Harnessing and utilizing solar energy is one of the most promising ways to reduce the mismatch between energy supply and demand [1]. Phase change materials (PCMs), as advanced thermal energy storage (TES) materials, are widely applied to storage energy due to their large latent heat and isothermal phase transition temperature during the phase change process. Consequently, solar energy acquisition and storage associated with PCMs have become a research hotspot [2–5]. However, visible light, which accounts for approximately 44% of solar radiation, cannot be directly or effectively applied in PCMs [6–9]. Thus, utilizing composite PCMs to harvest and convert visible sunlight into thermal energy has attracted significant attention in modern energy technology [1,3,4,10].

Currently, the materials that can absorb and convert light energy into thermal energy include mainly inorganic materials [11] and organic

pigments and dyes [12]. Inorganic materials are mostly applied in solar selective absorption coatings, such as carbon black [13–15], Cr_xO_y [16–18], and CuS [19,20], for their electron-jumping energy levels matching the photon energy in the near-infrared spectral region as well as their surface plasmon resonance effect [10]. They can also be doped into PCMs as photothermal conversion materials, but they are prone to uneven dispersion and phase separation. For example, incorporating CuS nanoparticles into the paraffin/ SiO_2 PCM can significantly improve the photothermal conversion performance of the composite PCMs [21]. On the other hand, organic photothermal conversion materials such as azo dyes, anthraquinone dyes, polyaniline, and indocyanine green (ICG) are easily connected to PCMs or support materials through chemical bonding and promote heat transfer performance, and avoid phase separation [22]. Besides, organic shape-stabilized PCMs with photothermal conversion and visible light-harvesting abilities can be achieved via condensation polymerization of the corresponding dye and PCM

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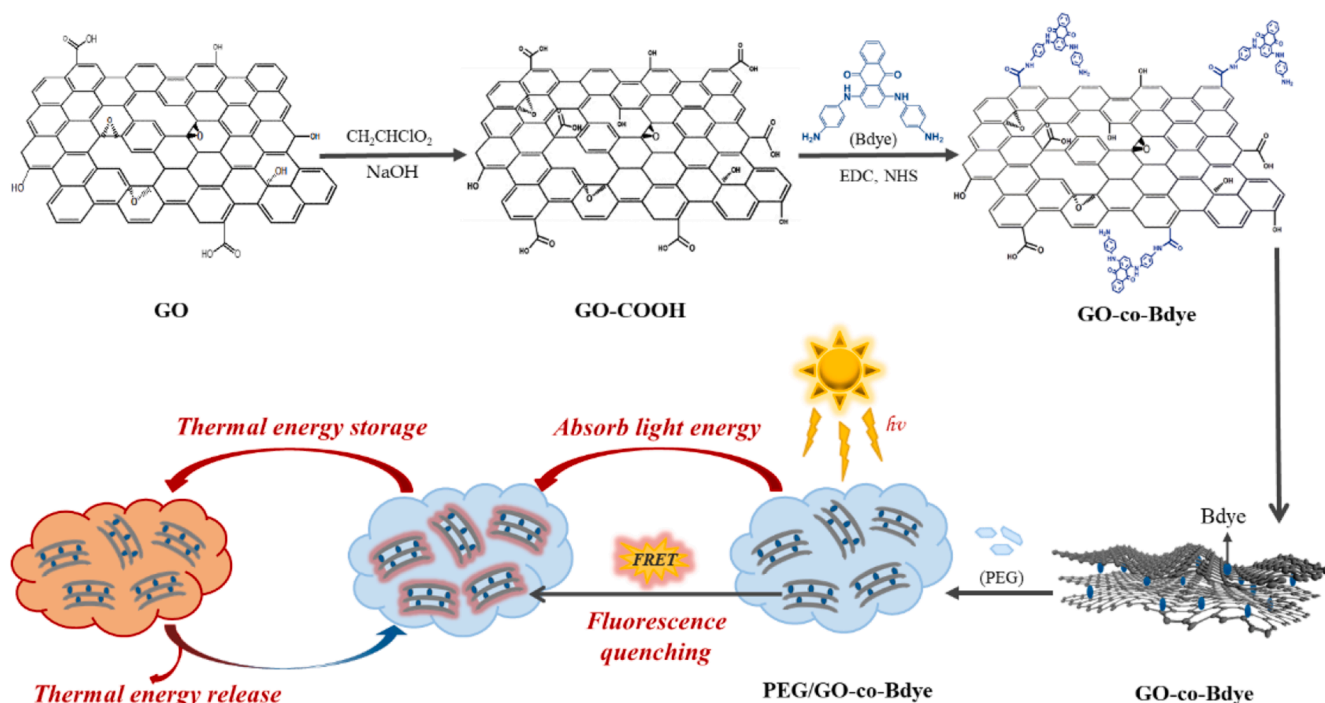


Fig. 1. Schematic illustration of the preparation process of PEG/GO-co-Bdye composite PCMs.

monomers. Dyes attached to the PCM monomers could harvest light and convert to heat, which is subsequently transferred to the PCM and stored [23]. Our group prepared shape-stabilized linear polycondensation photothermal conversion materials by condensing dye(phenolphthalein) containing two *iso*-reactive groups (hydroxyl groups), PEG, and diphenyl-methane-diisocyanate (MDI) through a two-step polymerization process [24]. The resulted material exhibited high strength and flexibility as well as superior photothermal energy storage and conversion efficiency. Due to sensitive non-radiative leapfrogging, organic pigments and dyes with conjugated structures are more inclined to convert the absorbed light energy into heat energy than inorganic materials [25]. Especially, anthraquinone dyes with strong visible light selective absorption and high molar extinction coefficients show excellent performance in the field of photothermal conversion [7]. However, the fluorescence produced by radiative leapfrog would still dissipate their photothermal conversion efficiency to some extent [8].

Herein, an intelligent photo-responsive blue anthraquinone-grafted GO nanoconjugate (GO-Bdye) system was developed, which can improve the photothermal conversion efficiency and reduce the fluorescence effect via exhibiting fluorescence resonance energy transfer (FRET) effect between blue anthraquinone dyes (Bdye) and GO (Fig. 1). GO with a large number of functional groups (carboxyl and hydroxyl) can be chemically modified and grafted with organic dyes and serve as supporting-material for PCM [8,26–29]. Bdye selectively absorbed 510 ~ 770 nm light radiation and converted light energy into heat energy mainly through non-radiative leapfrogging, which was stored by the PCM. Simultaneously, since the fluorescence emission spectrum of Bdye overlapped with the absorption spectrum of GO, the distance between the Bdye groups and GO sheets was extremely close due to the chemical bonding. The small amount of fluorescence generated by Bdye's radiative leap can transfer to GO, which acted as a fluorescent receptor and converted the received fluorescence energy to additional thermal energy, i.e., the FRET effect occurred. On the other hand, GO showed a large surface area, excellent mechanical properties, and regular structures [30–33]. Thereby, it improved the crystalline nature, thermal stability, and thermal conductivity of composite PCMs. The obtained shape-stabilized photothermal PCMs also displayed high PCM loading of

80.4 wt% and photothermal conversion efficiency of 86.6%, signifying that the composite materials are a promising candidate for photothermal management systems like solar water heaters and building insulation materials.

2. Experimental

2.1. Materials

Poly(ethylene glycol) (PEG) (molecular weight = 2000) was purchased from Sinopharm Chemical Reagent Co., Ltd and dried at 40 °C under high vacuum (0.1 kPa) for 48 h before use. Raw materials for the synthesis of blue anthraquinone dyes ($C_{26}H_{20}N_4O_2$, Bdye), Quinizarin ($C_{14}H_8O_4$) was produced by Tokyo Chemical Industry CO., Ltd. The catalysts for amide reaction, 1-(3-Dimethylaminopropyl)-3-ethyl carbodiimide hydrochloride (EDC) and 1-hydroxy pyrrolidine-2,5-dione (NHS) were purchased from Alfa Aesar (China) chemicals Co., Ltd. Sodium carbonate (Na_2CO_3) and N, N-Dimethylformamide (DMF) were obtained from Beijing Chemical Works. Other reagents were purchased from Guoyao, Inc., China, and used without further purification.

2.2. Synthesis of GO-COOH and GO-co-Bdye

GO was prepared by Hummer's method following previously reported literature [34]. The synthesis process of GO-COOH was carried out as follows: 1 g chloroacetic acid and 1.2 g sodium hydroxide were added into 100 ml GO suspension ($2 \text{ mg} \cdot \text{ml}^{-1}$) with the aid of ultrasonic bath treatment for 1 h, and then followed by magnetic stirring for 24 h to react sufficiently and form carboxylated GO [35]. Finally, GO-COOH was obtained by filtration, repeated rinsing with deionized water, and drying at 60 °C in a vacuum oven for 6 h.

The GO-co-Bdye was prepared by covalently grafting anthraquinone dye (the chemical structure was shown in Fig. 1) on the surface of GO via amide reaction between the amino groups of Bdye and carboxy groups of GO-COOH. 100 mg GO-COOH was dispersed into 150 ml anhydrous DMF and sonicated for 2 h. Then, 160 mg EDC was added and the solution was magnetically stirred for an hour. After adjusting the pH of the

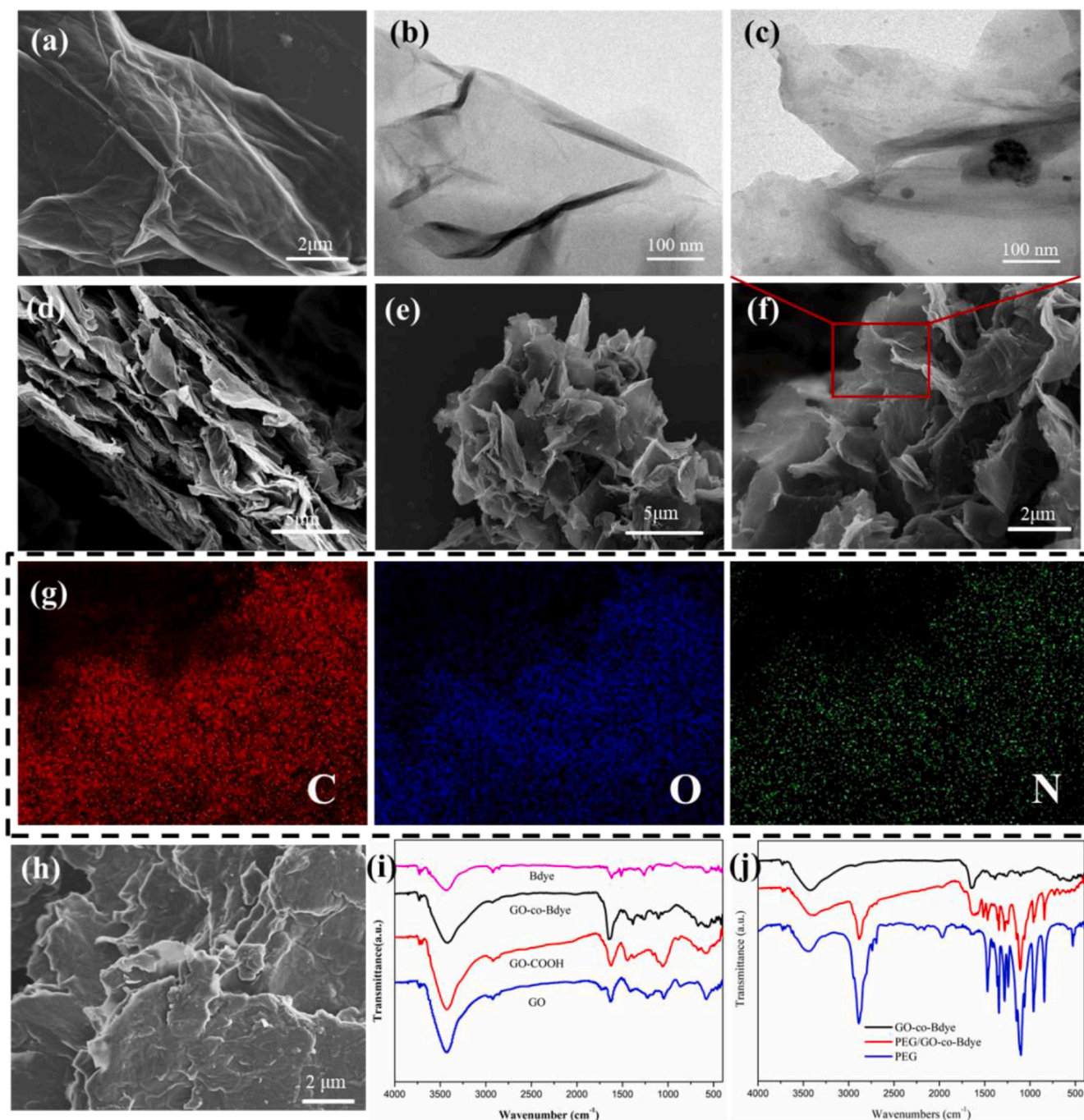


Fig. 2. (a and b) SEM and TEM image of GO nanosheets. (c) TEM image of GO-co-Bdye. (d-f) SEM images of GO-co-Bdye. (g) The EDS mapping images of GO-co-Bdye. (h) SEM image of PEG/GO-co-Bdye. (i and j) FT-IR spectra of as-prepared samples.

solution to 7 with triethylamine, 320 mg NHS was added and stirred for 30 min to form Sulfo-NHS ester with amino reaction activity. Then 120 mg Bdye was added to the solution and magnetically stirred for 12 h at room temperature. Finally, the product was obtained by filtration, washing 2–3 times with ethanol, and drying at 60 °C. As a comparison, a composite of GO and Bdye stabilized via π - π interactions (named as GO/Bdye) was also prepared by simply mixing the dispersible GO and Bdye.

2.3. Preparation of composite ss-PCMs

The composite PCMs, PEG/GO-co-Bdye, were prepared by a blending and impregnating method. GO-co-Bdye powder was dispersed in a flask with absolute ethanol with the aid of ultrasonic bath treatment for about

1 h. Then, the flask was transferred to an oil bath at 80 °C and stirred for 30 min. After that, 2 g of dried PEG2000 powder was added into the solution with magnetic stirring for 2 h until the PEG dissolved completely. Finally, the samples were dried at 80 °C in a vacuum oven for 8 h. PEG/GO/Bdye composite PCMs were synthesized in the same way.

2.4. Characterization

The morphologies of the as-prepared samples were characterized using scanning electron microscopy (SEM, ZEISS SUPRA55) operated at the accelerating voltages of 5 kV. The energy-dispersive X-ray spectroscopy (EDS) analysis were also measured by ZEISS SUPRA55.

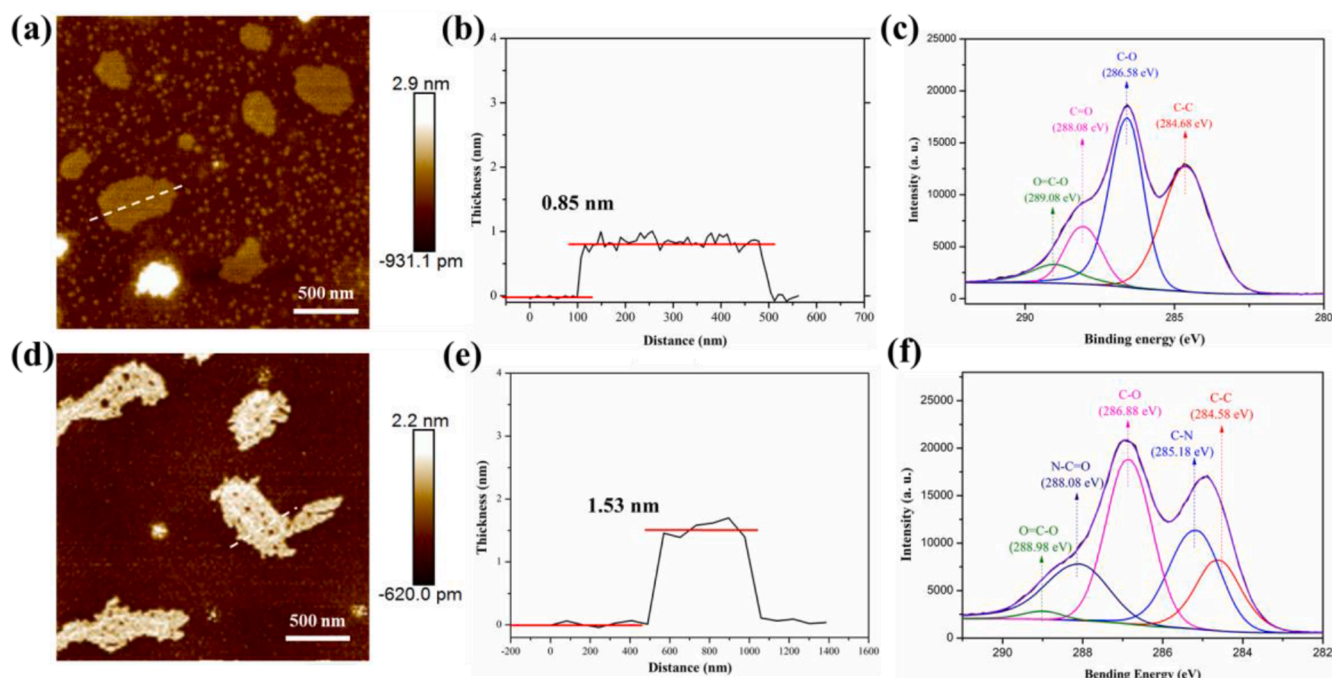


Fig. 3. AFM images of (a) GO and (d) GO-co-Bdye. The AFM height analysis of (b) GO and (e) GO-co-Bdye. The C1s XPS analysis of (c) GO and (f) GO-co-Bdye.

Transmission electron microscope (TEM) images were obtained by JEM-2010 operated at the accelerating voltages of 200 kV. Fourier-transform infrared spectra (FTIR) analysis was carried out by Nicolet 6700 spectrometer from wavelength ranges from 400 to 4000 cm^{-1} . Atomic force microscopy (AFM)-based single-molecule force spectroscopy (SMFS) was used to investigate the dimensions of the samples. AFM experiments were performed on a Nano Wizard II Bio AFM (JPK instrument AG, Berlin, Germany). The thermal stability of the samples was measured on a NETZSCH STA449F3 at a heating rate of 10 $^{\circ}\text{C}\cdot\text{min}^{-1}$ under a nitrogen atmosphere. Fluorescent microscopy images were recorded on a fluorescence spectrophotometer (Hitachi F-7000) at different temperatures. The thermal analysis (TGA) was used to investigate the thermal decomposition behavior of the samples measured on a NETZSCH STA449F3 at a heating rate of 10 $^{\circ}\text{C}\cdot\text{min}^{-1}$ under a nitrogen atmosphere. Phase change temperature and enthalpy of the composites were investigated using differential scanning calorimetry (DSC, Mettler DSC822e). The composites were heated and cooled between 5 $^{\circ}\text{C}$ and 80 $^{\circ}\text{C}$ at a rate of 5 $^{\circ}\text{C}\cdot\text{min}^{-1}$ under a nitrogen atmosphere. The crystallization process of the PEG in GO was recorded by temperature-dependent XRD measurements (Bruker D8 DISCOVER). All the samples were scanned from 5 $^{\circ}$ to 90 $^{\circ}$ at 5 $^{\circ}/\text{min}$, and Cu K α radiation ($\lambda = 1.541 \text{ \AA}$) was applied as an incident source.

3. Results and discussion

3.1. Morphological and structural analysis of samples

A schematic for synthesizing the novel PEG/GO-co-Bdye composite PCMs is represented in Fig. 1. GO was first treated with chloroacetic acid and sodium hydroxide to increase the amount of carboxyl group. Then the fluorescent dye (Bdye) was grafted onto the carboxylated GO surface via amide reaction. Strong covalent bond and affinity between the negatively charged carboxyl groups of GO sheets and the positively charged amino groups of Bdye molecules anchored Bdye tightly to GO sheets. Simultaneously, Bdye with two *iso*-active amino groups can connect GO nanosheets. As a result, GO-co-Bdye showed a 3D interconnected network structure, which facilitated the loading of the phase change material. GO-co-Bdye served as the photo-driven heating unit

that can effectively capture photons, quenching fluorescence, and convert light to heat. Subsequently, the heat was transferred from GO-co-Bdye to PEG, which underwent phase transition and achieved thermal energy storage.

As shown in Fig. 2a, GO nanosheets presented a large lamellar structure with folds and ripples, owing to deformation during spalling and re-stacking [36]. The TEM image of GO nanosheets also confirmed the lamellar structure of GO nanosheets (Fig. 2b). Whereas the GO-co-Bdye nano-composite showed a 3D “flower-like” interconnected network structure (Fig. 2d-f). Bdye connected with GO nanosheets via amide bonds distorted the two-dimensional (2D) planar structure of GO and further contributed to the formation of 3D structures. Compared with 2D GO, 3D GO-co-Bdye not only facilitated the sequestration of PCMs but also provided a continuous path for heat transfer. In addition, the microstructure of GO-co-Bdye revealed a uniform distribution of some small particles (Fig. 2c), which were attributed to the absorbed or grafted Bdye molecules. The uniform and compact distribution of Bdye molecules on the GO surface created a conducive environment for photo capture. Moreover, the energy-dispersive spectroscopy (EDS) mapping (Fig. 2g) of the N element showed the same morphology as those of the other elements, confirming the uniform dispersion of Bdye in this system. Also, PEG/GO-co-Bdye revealed the homogeneous absorption of PEG in the 3D network (Fig. 2h).

The FT-IR spectra of GO, GO-COOH, Bdye, and GO-co-Bdye demonstrated an amide reaction between the -COOH group of GO and the -NH $_2$ group of Bdye and the formation of amide bonds (Fig. 2i). In the spectra of GO, the band appearing at 3430 cm^{-1} corresponded to the stretching vibrations from the O-H group. The stretching vibration peaks at 1730 cm^{-1} and 1630 cm^{-1} were attributed to C=O in carboxyl and carbonyl groups. The peaks of 1127 cm^{-1} and 1230 cm^{-1} were assigned to the bending vibrations of C-O-C [37,38]. Compared with GO, the spectrum of GO-COOH showed a strong stretching vibration at 1621 cm^{-1} owing to the C=O group of the -COOH functional groups. The intense band at 1442 cm^{-1} corresponding to the characteristic absorption of the benzene ring. The characteristic peak of the ether band showed a redshift from 1050 cm^{-1} to 1058 cm^{-1} with weak peak intensity, indicating that chloroacetic acid transformed ether bands into carboxyl groups [39,40]. Similarly, Bdye showed N-H stretching

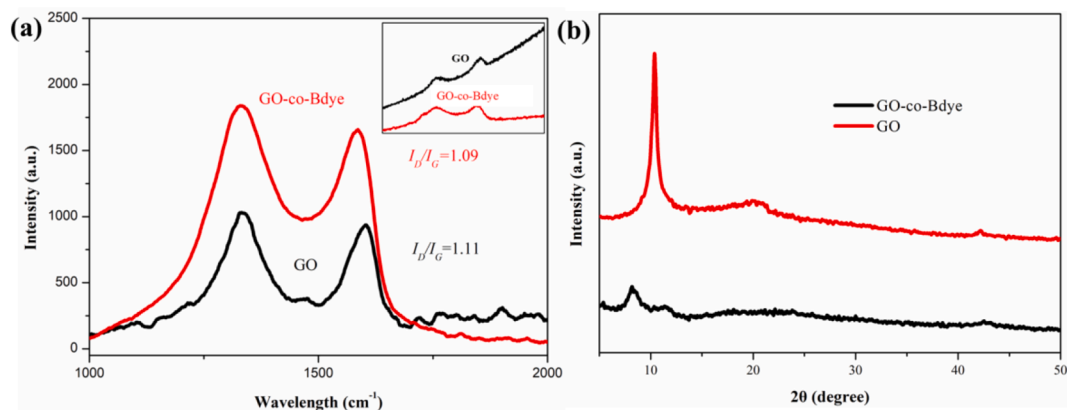


Fig. 4. (a) Raman spectra and (b) XRD diffraction patterns of GO and GO-co-Bdye.

vibration at 1617 cm⁻¹, while the peak at 1263 cm⁻¹ was ascribed to the bending vibration peaks. Comparing the spectra of GO-COOH and Bdye, the new stretching vibration peak appearing at 1635 cm⁻¹ could be attributed to amide carbonyl-stretching mode, and the peak at 1259 cm⁻¹ ascribed to the C-N stretching vibration or the N-H bending vibration [41,42]. Those characteristic peaks implied the existence of the amide band and the successful reaction between GO-COOH and Bdye.

The FT-IR spectra of the GO-co-Bdye, PEG, and their composite are shown in Fig. 2j. The strong peaks of PEG at 2888 cm⁻¹ and 1101 cm⁻¹ assigned to the stretching vibration of -CH₂-. The peaks at 3438 cm⁻¹ and 1639 cm⁻¹ were absorption peaks for O-H stretching vibration and C = O stretching vibration, respectively [2,43,44]. Moreover, PEG/GO-co-Bdye revealed the combined absorption peaks of pristine PEG and GO-co-Bdye with some slight offset peaks, indicating no chemical reaction between PEG and GO-co-Bdye.

Further, the AFM imaging demonstrated the lamella thickness after GO grafting with Bdye. The thickness of single-layer GO was 0.85 nm ± 0.02 nm (Fig. 3a and 3b). However, after the Bdye grafting, the thickness of monolayer GO-co-Bdye turned out to be over 1.53 nm (Fig. 3d and 3e), signifying the successful integration of Bdye and GO.

Meanwhile, XPS analysis was performed to investigate the elemental composition and chemical states of GO and GO-co-Bdye. The C 1s high-resolution XPS spectrum of GO can be deconvoluted into four peak components (Fig. 3c), which were associated with non-oxygenated ring carbon C = C (284.68 eV), epoxy and alkoxy carbon C-O (286.58 eV), and carbonyl carbon C = O or O = C-O (289.08 eV and 288.08 eV) functional groups [45,46]. Compared with GO, the XPS spectra of GO-co-Bdye illustrated that the values of the C-C (284.68 eV), C-O (286.58 eV), and O-C = O (289.08 eV) were decreased and the two new peaks of C-N (285.18 eV) and N-C = O (288.08 eV) emerged after modification (Fig. 3f).

The Raman spectra demonstrated two peaks located at 1344 cm⁻¹ corresponding to the D band (sp³ carbon structure of disordered graphitic layer), and 1599 cm⁻¹ related to the G band (sp² in-plane bonded carbons) (Fig. 4a) [47]. The relative intensity ratios I_D/I_G of the GO and GO-co-Bdye were 1.09 and 1.11, respectively, indicating that the samples exhibited a good graphitization degree ($r > 1.0$), which was favorable for heat transfer performance. Furthermore, it is noteworthy that the stronger the fluorescence effect of the material, the more the Raman spectrum was subjected to fluorescence interference, and the greater the tilt of the data line [48]. Interestingly, as shown in the inset of Fig. 4a, GO-co-Bdye exhibited a data line with a smaller inclination degree than GO before baseline correction, implying a reduced fluorescence effect of GO.

The crystalline morphology of GO and GO-co-Bdye were further studied by XRD (Fig. 4b). The GO synthesized by the modified Hummers method showed a strong diffraction peak at about 10.38°, which belonged to the (001) crystal plane [49]. However, the characteristic diffraction peak of GO-co-Bdye shifted left to about 8.16°, indicating that the carboxylation and Bdye grafting on GO enlarged the lamellar spacing.

3.2. Photo-to-thermal energy conversion

The UV-vis spectrum of the as-synthesized GO-co-Bdye, GO and pristine Bdye demonstrated that the characteristic absorption peaks of Bdye were located at 430 nm and 640 nm (Fig. 5a), respectively. The absorption curve of GO presented no characteristic absorption in the visible region but showed a relatively homogeneous absorption of light in the full wavelength band. The absorbance of GO-co-Bdye at 640 nm increased about 19 folds comparing with that of GO, signifying successful linkage of Bdye with GO. Compared to free Bdye, the

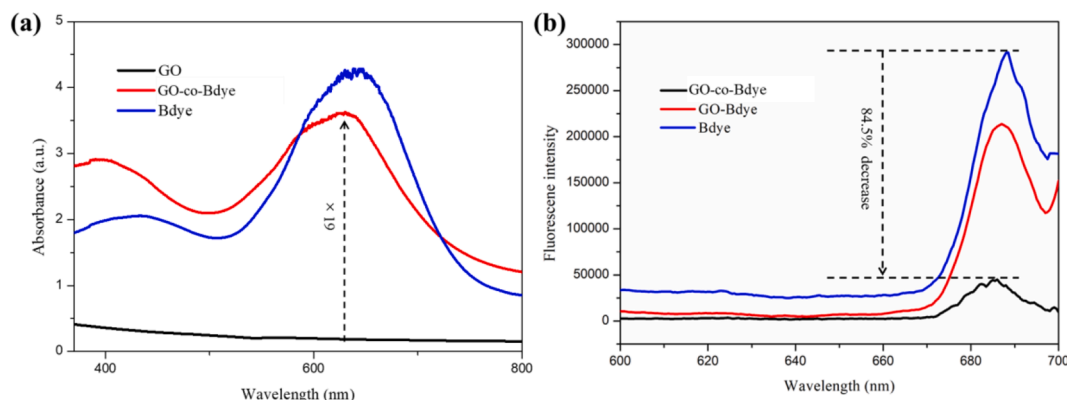


Fig. 5. (a) UV-vis absorption spectra of Bdye, GO, and GO-co-Bdye. (b) Fluorescence transient spectrogram of Bdye, GO-Bdye, and GO-co-Bdye.

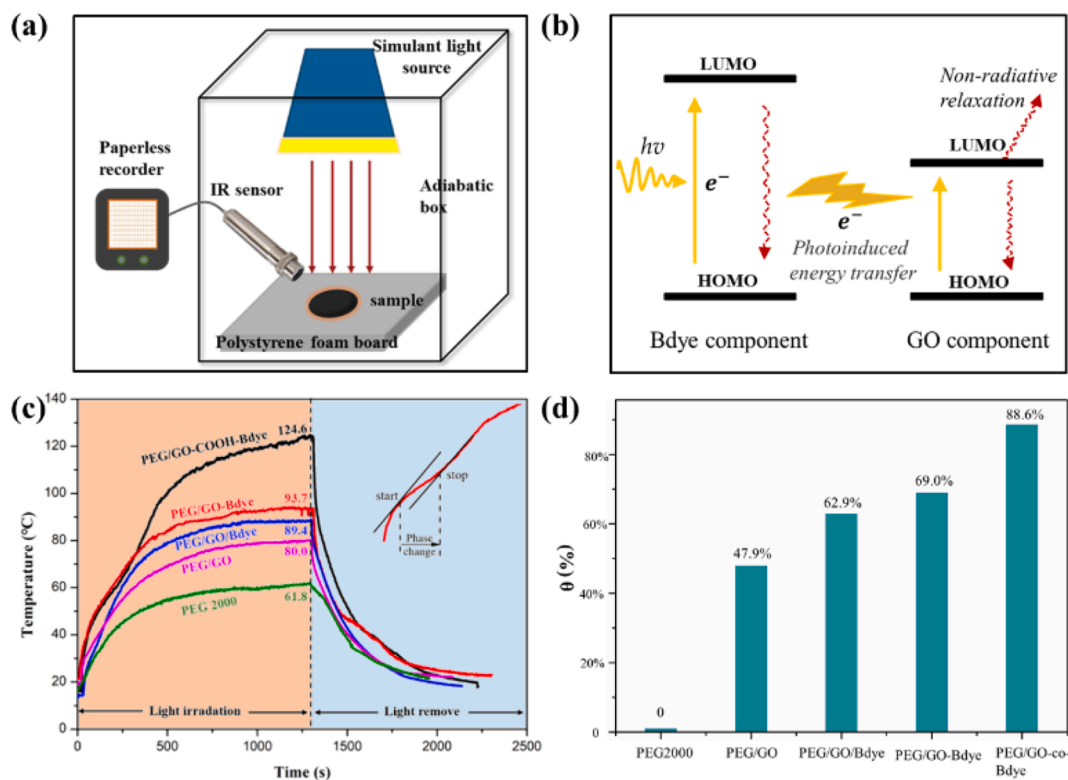


Fig. 6. (a) Schematic illustration of the photothermal performance measurement setup of the samples. The wafer samples of composites and PEG were inserted in a thermally insulating polystyrene foam board, and the surfaces of these samples with the same surface area were exposed to the simulated solar light. An IR sensor was used to read the temperature change of samples in real-time and transmit the data to the paperless recorder. (b) schematic illustration of the enhanced photothermal effect process (c) Temperature evolution curves of pure PEG, PEG/GO-Bdy, PEG/GO-co-Bdy composite PCMs under solar illumination. (d) The light-to-heat and energy storage efficiency (θ) of PEG, GO, PEG/GO-Bdy, and PEG/GO-co-Bdy.

fluorescence intensity of GO-co-Bdy at 687 nm decreased by 84.5% (Fig. 5b). The enhanced visible light absorption and decreased fluorescence emission of GO-co-Bdy were presumably due to the direct combination of Bdy and GO nanosheets. Such photoinduced electronic energy transfer from Bdy to GO is beneficial for translating the light to heat for thermal storage applications.

To assess the photothermal conversion ability of samples and determine the maximum temperature samples can reach under solar irradiation. The light-to-heat energy conversion measurements were performed as illustrated in Fig. 6a. Samples of the PEG/GO-co-Bdy, PEG/GO-Bdy, PEG/GO/Bdy, PEG/GO, and pristine PEG were irradiated under simulated solar light and a paperless recorder with an infrared (IR) sensor was applied to monitor the temperature change of the samples in air. To minimize the heat loss, a circular non-through hole was cut on the surface of thermally insulating polystyrene foam and the samples with the same weight of ~ 0.2 g were successively placed in the hole. The simulated solar light shined on top of the samples. All samples with the same top surface area of ~ 0.246 cm² were exposed to the light and the measurement was conducted at constant ambient temperature (~ 22 °C).

Fig. 6b shows the mechanism for the enhanced photothermal effect of PEG/GO-Bdy and PEG/GO-co-Bdy, which includes three processes: (1) absorption of light by Bdy, (2) photoinduced electronic energy transfer from Bdy to GO, and (3) dissipation of heat to the air via non-radiative relaxation in GO [50]. The photo-to-thermal energy conversion spectra of PEG/GO-co-Bdy composite PCMs were showed in Fig. 6c. Under the same solar irradiation condition, the temperature of PEG/GO-co-Bdy, PEG/GO-Bdy, PEG/GO, and pure PEG reached up to 124.6, 93.7, 89.4, and 61.8 °C, respectively. The maximum achievable temperature is a steady-state temperature in which the heat generation under light irradiation and heat dissipation due to radiative heat flux are

found at equilibrium. As a comparison, the PEG was also examined at elevated temperatures. However, the values were much lower than that of samples containing GO or Bdy. Typically, the temperature of PEG/GO-Bdy and PEG/GO-co-Bdy composite PCMs rapidly increased upon solar irradiation, which was ascribed to the photoinduced electronic energy transfer from Bdy to the GO framework by Bdy as an effective photon capture. Meanwhile, the PEG/GO-co-Bdy composite PCMs exhibited higher photothermal conversion capacity than that of PEG/GO-Bdy and PEG/GO, which indicated the dyes grafted on the carboxylation of GO can effectively facilitate the light-to-heat conversion and storage.

In addition, PEG/GO/Bdy with the same ratio of PEG, GO and Bdy was also prepared and investigated. Under the same light conditions, PEG/GO-co-Bdy showed a higher heating rate than PEG/GO/Bdy with the highest temperatures reached 124.6 and 89.4 °C, respectively. Such significant GO-co-Bdy improvement was due to the stable connection between GO and Bdy, which was in line with the IR results. The results indicated that the fluorescence resonance energy transfer from Bdy to GO played an important role in achieving an enhanced photothermal effect for the GO-Bdy composites.

When the temperature reached the melting point of the PEG/GO-co-Bdy, a phase change plateau was observed in the temperature evolution curves with a small fluctuation, signifying the occurrence of phase transition and thermal energy storage. Meanwhile, PEG/GO-co-Bdy absorbed solar radiation and converted it to thermal energy. After shutting off the illumination, the temperature of the PEG/GO-co-Bdy rapidly decreased and a cooling temperature platform emerged later as the PCMs changed from liquid to solid, releasing the thermal energy to maintain the temperature.

The light-to-heat and energy storage efficiency (θ) of the composite PCMs was evaluated following Eq. (1): [1].

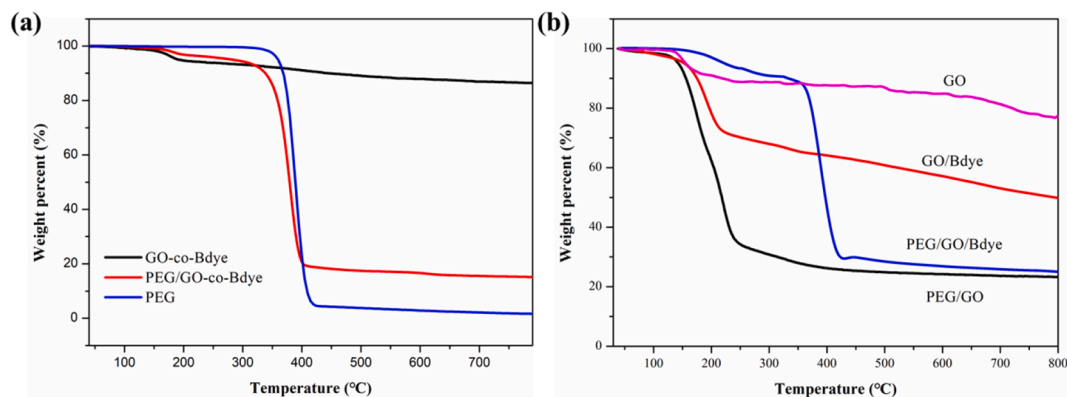


Fig. 7. TGA curves of (a) GO-co-Bdye, PEG, and PEG/GO-co-Bdye composite PCM, and (b) GO, GO/Bdye, PEG/GO/Bdye, and PEG/GO composite PCM.

Table 1

DSC heating and cooling characteristics of pure PEG and the composite PCMs.

samples	ω (wt %)	Melting process		Cooling process		ΔT (°C)
		T_m (°C)	ΔH_m (J·g ⁻¹)	T_c (°C)	ΔH_c (J·g ⁻¹)	
PEG	100	55.3	170.4	21.8	151.1	33.5
PEG/GO	68.6	59.5	96.9	27.5	84.0	32.0
PEG/GO/Bdye	69.1	54.5	109.4	30.7	122.1	26.4
PEG/GO-co-Bdye	80.4	53.7	130.6	31.9	127.1	21.8

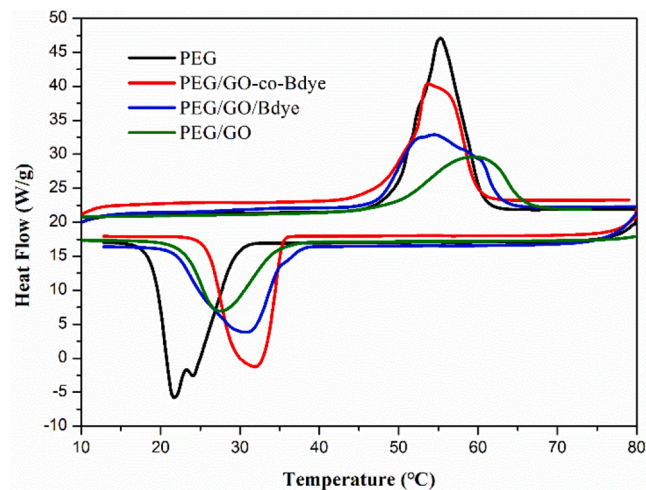


Fig. 9. DSC curves of pure PEG and the composite PCMs.

GO-co-Bdye shows potential fast-responsive photothermal energy storage and conversion performance.

3.3. Thermal properties

The thermal stability and thermal degradation behaviors of PEG, GO, PEG/GO, GO/Bdye, PEG/GO/Bdye, GO-co-Bdye, and PEG/GO-co-Bdye were investigated by TGA measurements (Fig. 7). The weight percentage (ω) of PEG in the PEG/GO, PEG/GO/Bdye, PEG/GO-co-Bdye calculated from the TGA curves were 68.6 wt%, 69.1 wt%, and 80.4 wt%, respectively, which were shown in Table 1. From the TGA curves in Fig. 7a, the weight decomposition of pure PEG occurred from 330 to 430 °C with a total weight loss of 96.0%. The initial decomposition temperature of GO-co-Bdye and PEG/GO-co-Bdye occurred at approximately 170 °C due to the evaporation of absorbed water. As to GO-co-Bdye, only a slight weight loss at high temperature occurred, corresponding to the reduction of unstable oxygen functional groups. For the PEG/GO-co-Bdye, the weight loss was almost similar to that of the weight loss of PEG, confirming that approximately 80.4 wt% PEG was loaded in GO-co-Bdye [51].

The X-ray diffraction patterns of PEG, GO-co-Bdye, and PEG/GO-co-Bdye revealed the crystalline morphology of the ss-PCMs (Fig. 8). Sharp and intense diffraction peaks at 18.96° and 22.80° indicated that PEG and ss-PCMs exhibited similar crystal structures and good crystallization properties. The characteristic peak of the support material in the PEG/GO-co-Bdye pattern migrated from 8.16° to 6.8° due to the expanded layer spacing of GO-co-Bdye by PEG, while the expanded space facilitated the mobility of PEG.

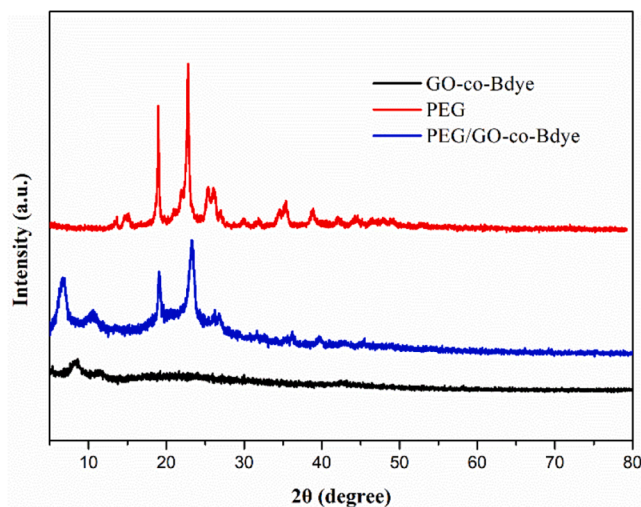


Fig. 8. XRD patterns of GO-co-Bdye, PEG, and PEG/GO-co-Bdye composite PCM.

$$\theta = \frac{m\Delta H}{\rho S(t_i - t_f)} \quad (1)$$

where m is the quality of the composite PCMs, ΔH is the transition enthalpy of the composite PCMs obtained by DSC, t_i and t_f are the photo-driven phase transition time of the respectively, ρ is the intensity for the light irradiation of the simulated light source, and S is the surface area of the samples.

The calculated θ values of pristine PEG and as-prepared composite materials are summarized in Fig. 6d. As expected, the PEG/GO-co-Bdye showed a superior θ value of approximately 88.6%, which was ~2 times higher than that of PEG/GO simulated under similar solar illumination intensity (70 mW·cm⁻²). GO-co-Bdye. These results indicated that PEG/

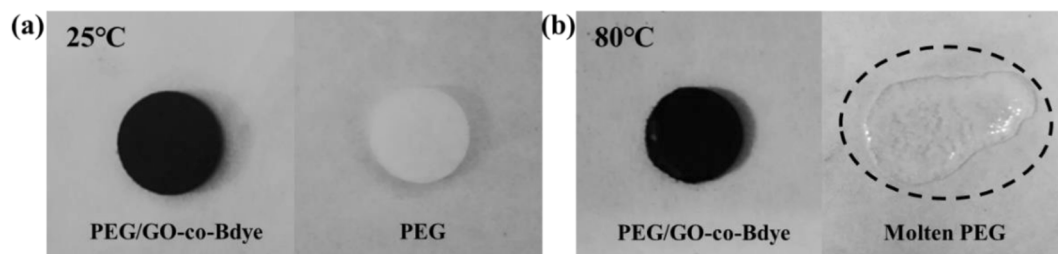


Fig. 10. The Shape-stabilized performance of the samples (a) at room temperature (25 °C) and (b) above the melting temperature (80 °C).

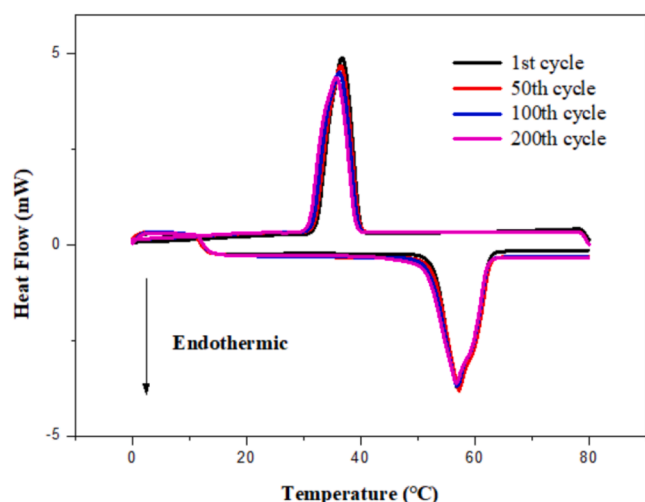


Fig. 11. DSC thermograms for the thermal cycling stability of the PEG/GO-co-Bdye composite PCMs over 200 thermal cycles.

The phase change behaviors of PEG and the composite ss-PCMs with different PEG loading (ω) were characterized via DSC (Fig. 9). The phase change enthalpies of the samples were calculated based on the area under the DSC curve during the melting/crystallization process. Based on the DSC analysis, the melting temperature (T_m), cooling temperature (T_c), melting latent heat (ΔH_m), cooling latent heat (ΔH_c), and undercooling degree ($\Delta T = T_m - T_c$) of shape stable PEG/GO, PEG/GO/Bdye, PEG/GO-co-Bdye composites and pure PEG were presented in Table 1.

The melting temperature of PEG/GO-co-Bdye was similar to that of pure PEG, while crystallization temperature of PEG/GO-co-Bdye was higher during the cooling process. As a result, the undercooling degree of pure PEG became decreased. The PEG/GO-co-Bdye exhibited the lowest undercooling degree ($\Delta T = 21.8$ °C) due to the 3D interconnected heat transfer path in the GO-co-Bdye. Besides, the theoretical latent heat (ΔH_{theo}) of the ss-PCMs was evaluated using Eq. (2): [52].

$$\Delta H_{theo} = \Delta H_{PEG} \times \omega \quad (2)$$

where ΔH_{PEG} represents the latent heat of pure PEG and ω refers to the weight percentage of PEG in the ss-PCMs calculated from the TGA results. The ΔH_m of pure PEG was $170.4 \text{ J} \cdot \text{g}^{-1}$ and the ω was 80.4 wt%. Therefore, the ΔH_{theo} of PEG/GO-co-Bdye was reached $139.9 \text{ J} \cdot \text{g}^{-1}$, which was similar to the experimental data recorded, $130.6 \text{ J} \cdot \text{g}^{-1}$. Compared with pure PEG, the composite PCMs exhibited a partial loss in latent heat because some mass fraction of PEG was replaced by supporting materials. Nevertheless, the loss was reduced with the increasing proportion of dyes grafted on GO probably due to the hydrogen bonding and $\pi \cdots \pi$ interaction between the Bdye and PEG. The T_c of the pristine PEG and PEG/GO-co-Bdye composite PCMs were quite close, indicating that the infiltration of the PEG in the 3D network showed little effect on its thermal properties.

Table 2

Thermal properties of the PEG/GO-co-Bdye composite PCMs at different thermal cycling tests.

Cycle times	Melting process		Freezing process	
	T_m (°C)	ΔH_m (J·g ⁻¹)	T_f (°C)	ΔH_f (J·g ⁻¹)
1	56.1	130.6	37.4	128.4
50	56.4	126.8	37.0	125.0
100	55.9	125.4	36.9	123.7
200	56.5	125.3	36.6	123.6

3.4. Shape-stability and thermal reliability analysis for composite PCMs

Shape-stability is a crucial characteristic for the prevention of leakage, storage, and transport of ss-PCMs. The shape-stability of pristine PEG and PEG/GO-co-Bdye was characterized through a leakage test by setting samples in an oven at different applied temperatures. As showed in Fig. 10, all the samples were solid with defined shapes below 55 °C (the melting point of PEG2000). The PEG/GO-co-Bdye maintained its original shape without significant melting-distortion and leakage of the liquid even heated above 80 °C, while the pristine PEG pellet quickly melted within 5 min. It revealed that GO-co-Bdye restricted the movement of PEG molecules and prevented leakage during the melting process. This is ascribed to the existence of hydrogen bonding and capillary force between GO-co-Bdye sheets and PEG molecules.

Thermal reliability is one of the critical parameters for the utilization of composite PCMs. Therefore, the thermal reliability of PEG/GO-co-Bdye was conducted after repeated melting-freezing cycles such as 1 cycle, 50 cycles, 100 cycles, and 200 cycles (Fig. 11 and Table 2). After 50, 100, and 200 thermal cycles, no obvious changes in either the exothermal or endothermal curves were observed. Although the crystalline peaks shifted slightly to the low-temperature region after thermal cycling, the crystallization temperature was still around 37 °C.

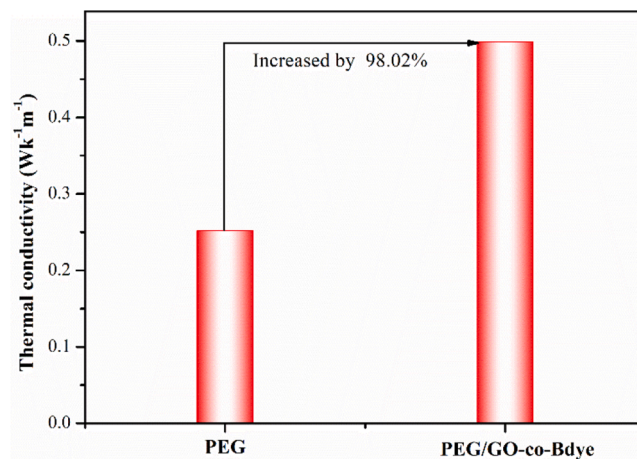


Fig. 12. The thermal conductivity of pristine PEG and PEG/GO-co-Bdye composite PCMs.

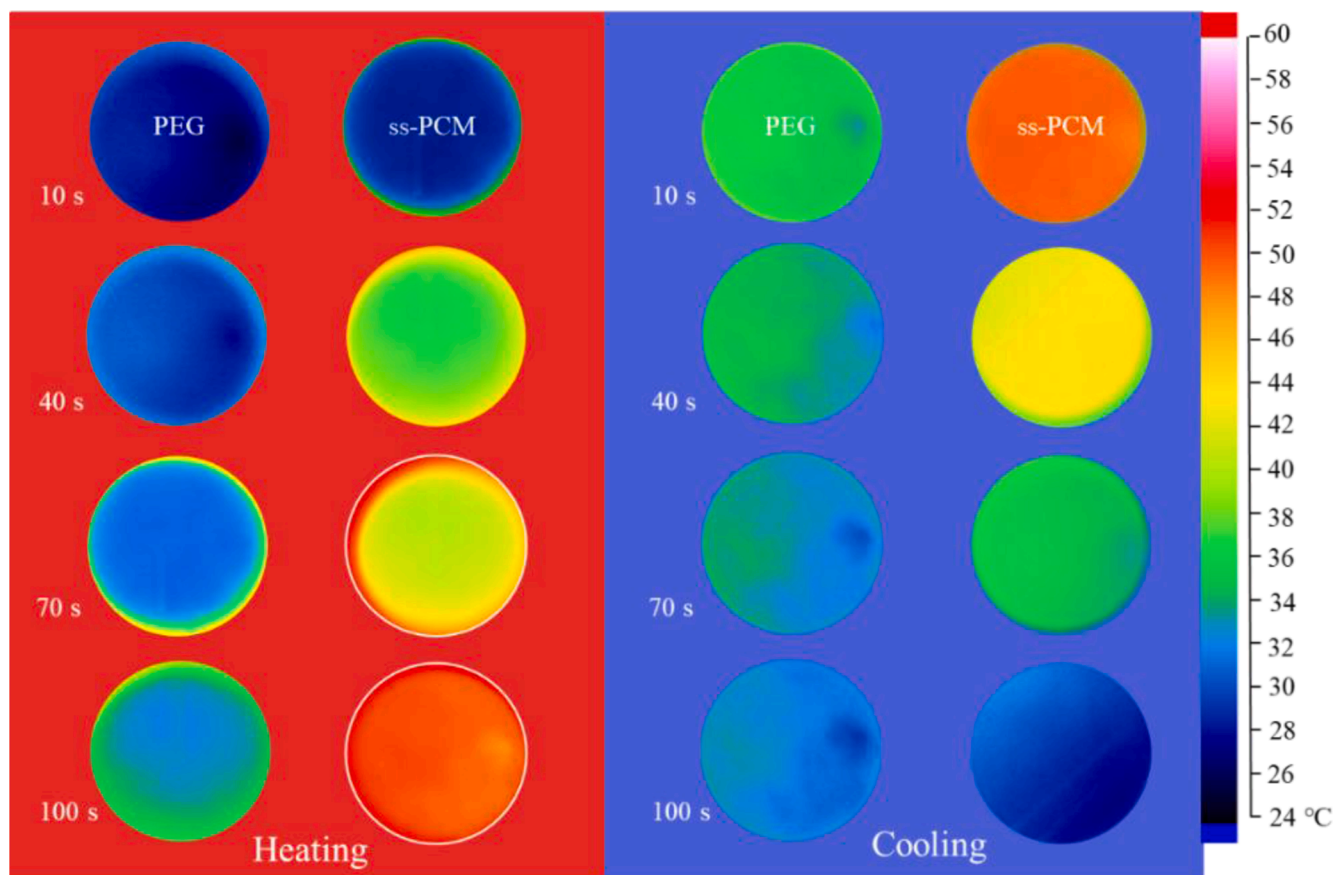


Fig. 13. Thermal images of PEG (left) and PEG/GO-co-Bdye (right) composite PCM during heating and cooling for 10, 40, 70, 100 s.

Meanwhile, the latent heat value showed a slight change after 200 thermal cycles. The ΔH_m of the PEG/GO-co-Bdye after 50, 100, and 200 thermal cycles decreased by only 2.9%, 3.9%, 4.0%, and the ΔH_f changed by only 2.6%, 3.6%, 3.7%, respectively. Therefore, the synthesized PCMs are suggested for large-scale practical applications such as building insulation and solar energy storage systems.

3.5. Thermal conductivity

Thermal conductivity is an important parameter to evaluate the thermal transport from the source to its nearby environment, which has a great impact on the photothermal conversion and thermal storage [6]. GO exhibited high thermal conductivity due to large grain size and strong carbon-carbon covalent bonding. Although the thermal conductivity of GO-co-Bdye showed a slight decrease due to the surface groups disrupting the structural integrity, it is still much higher than that of PEG. As a result, the thermal conductivity of PEG/GO-co-Bdye composite PCM was $0.499 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, which is 98.0% higher than that of pristine PEG ($0.252 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$) (Fig. 12). The close contact between the GO-co-Bdye and PEG in the composite enabled the composite to inherit the high heat transportability of the GO-co-Bdye and provided better heat transfer pathways.

The temperature response behaviors of samples were further examined via an infrared camera, which is essential to evaluate the heat response capacities and thermal transfer rates of PEG and ss-PCMs. The temperature distribution images during the heating and cooling process are shown in Fig. 13. During the heating process, the color change of the two samples implied that the temperature response rate of PEG/GO-co-Bdye was much faster than that of pure PEG because of the supporting material that markedly improved the thermal conductivity of the ss-PCMs. As expected, the surface temperature of the ss-PCMs

continuously declined at a higher rate during the cooling procedure. The results also illustrated that the novel PEG/GO-co-Bdye composite contributes a remarkable role in thermal energy management systems.

4. Conclusions

In this work, PEG/GO-co-Bdye capable of photo-driven energy conversion and storage was designed and synthesized via grafting blue anthraquinone dyes (Bdye) on carboxylated GO and impregnating PEG into GO-co-Bdye 3D network supporting materials. The composites effectively stored thermal energy through phase transition triggered by solar-to-thermal energy conversion under solar illumination with a high energy storage efficiency of 88.6%. GO sheets modified with anthraquinone dyes (GO-co-Bdye) exhibited an enhanced photothermal effect compared with pristine Bdye and materials designed with physically mixed ones. Therefore, the high fluorescence resonance energy was transferred from Bdye to GO sheets in the GO-co-Bdye scaffold. Moreover, the PEG supported with the 3D interconnected network structure of GO-co-Bdye showed improved thermal conductivity of $0.499 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, which was 98.0% higher than that of pristine PEG. In addition, the obtained PCMs exhibited high energy storage densities, good durability, and leakage-resistance capability, indicating their profound potential in large-scale energy harvesting, conversion, and storage applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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